

# Building a unified science of sensorimotor control with Densely Observable Measurement Environments [DOMEs]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

## 1. Mission

The MISSION of the **FreeMoCap Foundation X-Lab (FMC-X)** is to develop and disseminate the tools to measure a new scientific observable representing the closed loop of the **perceptual, mechanical, and environmental** aspects of goal-directed behavior of freely moving organisms to reshape perceptual/motor neuroscience, musculoskeletal biomechanics, embodied/agentive artificial intelligence, and mobile robotics into a new convergent science of **sensorimotor control in natural environments**.

### 1.1. Flagship

We will develop and validate a **novel research instrument** we call a **DOME** (*Densely Overlapping Measurement Environment, Ref - Figure 1 - Hero Figure showing full DOME*), which unifies a heterogeneous array of multi-modal sensors into single, centrally-controllable, metrologically-validated instrument for the measurement and manipulation of a freely moving agent within a specified 3D volume of real-world space (ref DOME-L, DOME-S, and DOME-Mobile in 3. Outcomes).

### 1.2. Unmet institutional needs

- Conway's Law
  - Academia too siloed for true cross-disciplinary works
  - Funding mechanisms defined as single-discipline institutions. Impossible to properly target proposals to announcements
  - Rome-on-top-of-Rome development of academic tools defines complexity ceiling for research tools (SKP-EI to help build to global scale)
- System rewards novel discoveries, not long slow work of proper tool making (Tools should be built by masters and taught to students, not other way around)

We will organize our technical development around the technical bottlenecks which limit our ability to combine reconstructions the **eye, body, and environment** into combined representations which trace to activity in the central and peripheral nervous system ...

- Best Eye trackers ever made at any price
- Best Mocap system ever made at any price
  - (but we'll sell em **real** cheap)
- Completely new fully integrated platform, exposing entirely new observable defined by densely overlapping measurements from the DOME
- DOMEs **everywhere**
  - Every human or animal-focused lab
  - Every hospital and PT clinic
  - DOMES on the space station and Vomit Comet rocket (study sensorimotor control and VOR in microgravity)
  - DOMES in athletic centers - Training athletes for high performance through education into their underlying sensorimotor control systems

- ▶ DOMES in mobile robot and drone labs - Align with human research, stop the robot folks from handrolling their own mocap all the time
- ▶ DOMES at the bottom of swimming pools and integrated into SCUBA systems of underwater welders (and other specialized trades)
- ▶ Mobile DOMES studying animals in natural environment, mounted to the bottoms of ships and integrated with ultrasonic sensors to mocap dolphins playing in the wake
- ▶ Tiny DOMEs to study honeybee and dragonfly visual flight control
- ▶ DOMES in every classroom!
  - In both science/tech/health classrooms AND animation/game design classrooms (DOMEs for animation mocap, freemocap already proved this works)
  - Always build “cheaper than a textbook” version of all hardware
  - Comensurate data output (with less precision and fewer modalities)
  - Prepare new generation for a future where human and robot centered research is focused on full-body kinematics and 3d binocular eye tracking with retinal projection.
  -



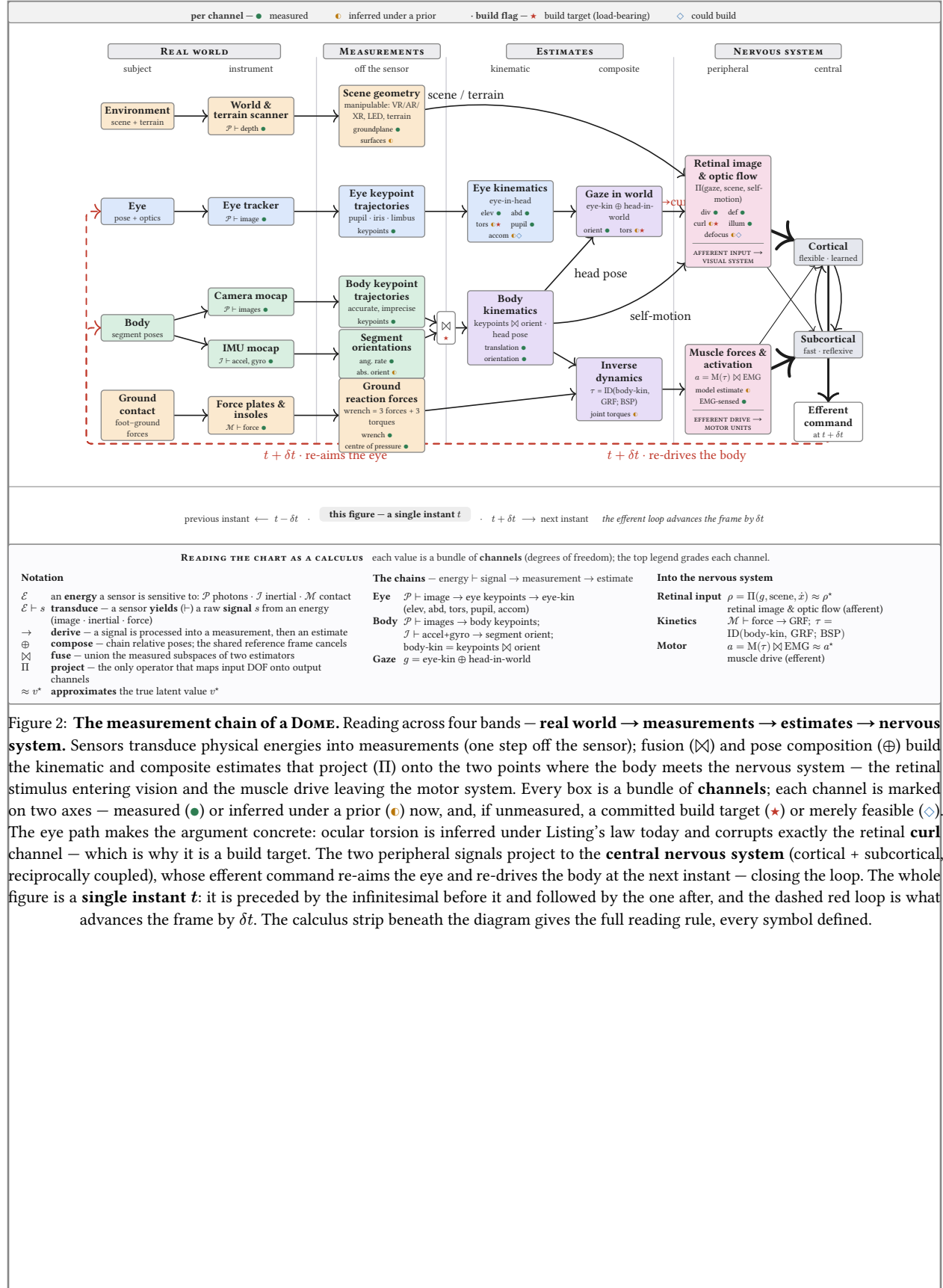


Figure 2: **The measurement chain of a DOME.** Reading across four bands — **real world** → **measurements** → **estimates** → **nervous system**. Sensors transduce physical energies into measurements (one step off the sensor); fusion ( $\otimes$ ) and pose composition ( $\oplus$ ) build the kinematic and composite estimates that project ( $\Pi$ ) onto the two points where the body meets the nervous system — the retinal stimulus entering vision and the muscle drive leaving the motor system. Every box is a bundle of **channels**; each channel is marked on two axes — measured (●) or inferred under a prior (◐) now, and, if unmeasured, a committed build target (★) or merely feasible (◇). The eye path makes the argument concrete: ocular torsion is inferred under Listing's law today and corrupts exactly the retinal **curl** channel — which is why it is a build target. The two peripheral signals project to the **central nervous system** (cortical + subcortical, reciprocally coupled), whose efferent command re-aims the eye and re-drives the body at the next instant — closing the loop. The whole figure is a **single instant**  $t$ : it is preceded by the infinitesimal before it and followed by the one after, and the dashed red loop is what advances the frame by  $\delta t$ . The calculus strip beneath the diagram gives the full reading rule, every symbol defined.

## 2. Technology Landscape

- Existing tools and methods
  - Some neat softwares and CV/ML tools, but all single threads, no unified tool
    - OpenSim - Widely used, very difficult to operate, highly specialized niche within biomechanics, single-modality
    - DeepLabCut - Widely used, very difficult to operate, single-modality
    - OpenCap works, but only for iPhones and requires a cloud upload, and caters exclusively to clinical biomechanics. Single-modality, non-convivial.
    - SIMPL/Meshcapade - Has built past empirically validity. 3d from 2d and mesh-fitting pipelines not truth preserving. Building increasingly more abstract and less tracable data channels on top of the same 2d data we've had for a while
    - etc
    - MuJoCo/IsaacLab- Driving revolution in mobile robotics! We will directly contribute to this by feeding RL robots DOME data, and then using resulting control theory to design controlled experiments (see ARGPv3)
- FreeMoCap and its landscape
  - Established Tool with Growing International Community
    - >15k uses from 152 countries, 10k GH stars, 3.5k in Discord server
  - Existence is proof-of-concept of approach, XLabs scale support would accelerate FreeMoCap model into the global scale
  - v2 has realtime and professional architecture (thanks to EI)
  - DOME-L and DOME-S camera-based mocap will use FMC or a variant. Mocap, Eye, World camera software composable from FMC polyrepo (skellycam, skellytracker, skellyforge, etc)
- ARGP
  - ARGPv1 - PhD Era<sup>1-4</sup>
  - ARGPv2 - NEU Era (Fuzzed out due to complexity, frustration spawned freemocap)
  - DOME includes ARGPv3
- Laser Skeleton research
  - <sup>5-7</sup>
  - etc
  - Ferret work
  -
- Existing Instruments
  - **Modern mobile eye trackers suck**
    - Pupils labs - Used to be cool, black box proprietary now. Best Available, used in Ferret and Laser Skeleton work.
    - Tobii et al not worth mentioning
    - Missing DoF's!
      - Can't measure Torsion
        - Field has invoked Listing's Law to ignore torsion for decades
        - Listing's Law is NOT APPLICABLE when VOR is active
        - VOR is active during 100% of natural behavior

- ▶ Torsion directly affects axis that Retinal Curl<sup>6</sup> lives on, so inability to measure it corrupts signal we want to propagate into visual cortex
- ▶ We know how to measure Torsion (iris texture), and Lens Accommodation too, while we're at it (yates et al DPI)

- ▶ **Modern mocap sucks**

- Old, awful, expensive, black boxes
- So difficult to use it's a literal bottleneck
- Difficult limits lab size (Can't re-target cameras without shutting down the lab for a week. Can't maintain multiple configurations)
- Theia exists, but it's re-building the same problems of the previous generation in a new modality

**3. Outcomes**

- Develop, validate, and disseminate - DOME Variants
  - **DOME-L** — the flagship instrumented volume, in the greater Boston area. Large enough to fully enclose the smaller variants for validation and metrological grounding.
  - **DOME-S** — the extension of the commodity-camera capture volumes already built and disseminated through the FreeMoCap Project. Representative of the DOME a research lab or a classroom would build.
  - **DOME-Mobile** — a wearable, self-contained DOME producing Dense Observations in unconstrained indoor and outdoor environments, continuing the lineage of the PI's gaze/gait and retinal optic flow work<sup>5-8</sup>.
  - Also - disseminate custom use-directed builds of DOME variants through network of collaborators studying Humans, animals, and robots.
- New eye tracker
  - Best ever made
  - Add every sensor we can, see which allow us to measure DoFs we need, optionally scale back and remove
  - Optionally also make cheaper version for classrooms - fully compatible data output (with less precision and depth)
    - Remake the field by retraining the next generation of researcher in the tools of the field you want to build
  - Dense instrumentation in the eye and world cameras
    - **Eye sensor array**
      - includes
        - High res IRGB
        - High speed IRGB
        - MEMS single-pixel sensor
        - LiDAR scanner
        - IR/Structured IR illuminators
        - Event Camera
    - **World sensor array**
      - Build different form factors (big heavy, small light)
      - Head- or Drone- mountable
      - Stereo RGB, Structured IR, LiDAR, etc
      - High density multi-model environment scan
      - Fast scan (RGB-D)
      - Slow scan (Textured Mesh)
      - Semantic segmentation (via SAM3, YOLO, etc)

|               |           |                                                                                                                                                                                                                                                                                                                                     |
|---------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.1. Timeline |           |                                                                                                                                                                                                                                                                                                                                     |
| Phase         | Name      | Description                                                                                                                                                                                                                                                                                                                         |
| 1             | Organize  | Set up Organization (Phase 0) – Solidify release of FreeMoCap v2                                                                                                                                                                                                                                                                    |
| 2             | Architect | Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research). Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems)                  |
| 3             | Execute   | Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3). Iterative design of instrumented system                                                                                                                                                                                        |
| 4             | Stabilize | Establish OODA loop within expected standard operations                                                                                                                                                                                                                                                                             |
| 5             | Expand    | Once stabilized, expand key performance metrics                                                                                                                                                                                                                                                                                     |
| 6             | Align     | Focus on building deeply integrated community of users and contributors (Transition BDFL → BDFaSPoT)                                                                                                                                                                                                                                |
| 7             | Establish | Focus on long term stability and healthy growth                                                                                                                                                                                                                                                                                     |
| 8             | Maintain  | Beyond proposal timeline – Establish organization practice in service of unbounded infinite timeline. Apply winning model to support extending ‘mesoscale research revolution’ to all biological sciences (e.g. integrating deeper microbio/biochem insights beyond the scope of the present perceptuomotor/biomech/robotics focus) |

Table 1: Timeline of Phased Milestones



4. Senior/Key Personnel Qualifications

4.1. Jonathan Matthis, PhD [Principle Investigator]

4.2. EI

4.3. AC,PhD

4.4. RR, CPA or w/e

| Name / Role                        | Qualifications                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jonathan Matthis, PhD<br>President | <ul style="list-style-type: none"><li>• Founder and chief maintainer of FreeMoCap project</li><li>• Left tenure track academic research position precisely because the institution could not support the work that needed to be done</li><li>• Decades experience integrating visual neuroscience with biomechanics/robotics<sup>5-7</sup></li></ul>                                                                                      |
| NR,<br>CEO                         | <ul style="list-style-type: none"><li>• Expertise in technically and software dev</li><li>• Previous CEO XP, founded and ran own buisness</li><li>• CEO helps PI apply his specific expertise to the core tasks of the XLab</li></ul>                                                                                                                                                                                                     |
| EI<br>CTO                          | <ul style="list-style-type: none"><li>• Decades of experience in software Industry</li><li>• Chief architect and CTO for established companies</li><li>• Represents expertise truly absent from academic contexts</li><li>• Ensures world-class enterprise-level software capability</li><li>• Will build training/workshop/hackathon backend to inject EI-level software development expertise into broad community of science</li></ul> |
| AC, PhD<br>CSO                     | <ul style="list-style-type: none"><li>• Expertise in validation of clinical research tools and bench-to-bedside development</li><li>• [CITE DISSERTATION]</li><li>• Experience in neuroprosthetics design and brain imaging</li><li>• Worked with PI on core FMC dev and validation</li></ul>                                                                                                                                             |
| JKL, PhD<br>CAIO                   | <ul style="list-style-type: none"><li>• PhD - Cognitive Science</li><li>• Director - Cognitive Science Program</li><li>• Current - Senior AI Applications Engineer @ SOLID</li><li>• Expertise in use of AI in classroom, lab, and enterprise</li></ul>                                                                                                                                                                                   |
| RR<br>CFO                          | <ul style="list-style-type: none"><li>• Bookkeeping</li><li>• NSF Grant management</li><li>• Finances</li><li>• Entrepreneurship (founded own Bookkeeping coop)</li><li>• Taught Finance Theory at university level</li></ul>                                                                                                                                                                                                             |
| KM<br>Project Manager (?)          | <ul style="list-style-type: none"><li>• Key driver of<sup>7</sup></li><li>• Worked with PI on core tech<sup>6</sup></li><li>• Expertise in key technologies</li></ul>                                                                                                                                                                                                                                                                     |
| MN<br>Project Manager (?)          | <ul style="list-style-type: none"><li>• Expertise in clinical biomechanics,</li><li>• Medical systems design,</li><li>• Mechatronics for clinical tools,</li><li>• Multi-modal motion capture lab management</li></ul>                                                                                                                                                                                                                    |

Table 2: Senior/Key Personnel Qualifications

## 5. Team Capabilities Statement

### 5.1. Senior Team Capabilities

- Jon - done it before in humans, currently doing it in ferrets and mice. Left tenure track position to run FreeMoCap, XLab funding will accelerate that mission
- EI - Deep professional experience in Tech Sector, high level software architecture experienced necessary to develop truly global scale enterprise level of software in open-source/academic contexts
- RR - Organizational health and entrepreneurial development
- NR - XP as CEO and working in tech industry. Hiring her as CEO allows Jon/PI to focus on applying their domain expertise on the core work of the lab
- JKL - AI expertise in teaching, research, and industry. Will build internal AI systems for core software, developer, user, and student assistance.
- AC - focusing clinical adoption and bench-to-bedside transitions. Wrote his dissertation on validation of freemocap. Has all skillsets needed for DOME-S
- MN - XP in mechatronics and large-scale lab building and management. Has core skillset needed for DOME-L
- KM - Part of original laser skeleton team, has all necessary sub-skills to build core instrumentation of Mobile DOME

### 5.2. Collaborators

| Initials           | Model / Domain       | Expertise                                  |
|--------------------|----------------------|--------------------------------------------|
| BS                 | Ferrets              | Visual/Perceptual neuroscience, ephys      |
| MD                 | Guinea Fowl          | Musculoskeletal Biomechanics, muscle units |
| DF                 | Mice                 | System Biology                             |
| AH                 | Marmosets / NHP      | Ephys                                      |
| MH                 | Human                | Vision and CPS                             |
| Mayo Clinic        | Human                | Brain scans and implanted sensors          |
| BD, GN/RAI, CH, JH | Robots / Fabrication | Control theory                             |
| AA                 | Physical facilities  | Rapid iteration, recruiting                |

Table 3: Collaborator Network

### 5.3. Governance

- FMC-X will exist as a independent project within FMC-F organization
- FMC-F will handle other work (e.g. FreeMoCap Project), but otherwise the missions almost fully overlap
- Nearly identical leadership between FMC-F and FMC-X, with final say on all topics falling to the President (who is also the PI) ensures fast decision making (no higher management to talk to - PI is the top of the org chart)
- Collaborator network and user community consulted for design decisions, work targets, and roadmap
  - Handled via “Request for Comments” style announcements (like Python’s PEP), so progress can’t be stalled by community problems.
- Annual Workshop/Hackathon/Conference/Congress to bring together collaborator network to discuss progress, tech targets, friction, etc.

- Workshops to train students and out-of-domain experts
- Hackathons to on-board developers and foster in-person connections between normally remote workers (important for FOSS)
- Conferences to discuss progress and research done with DOME tech
- Congress to discuss org status, plans, and future (annual State of the Skelly address)
- Developer fund, Community Grants Program, etc to support FOSS network
  - Use Community Grants Program to test different strategies for community engagement and consensus taking (Eurovision style voting, DAO methods, etc)

#### 5.4. Autonomy

- We are already autonomous, as of 2 weeks prior to the deadline of this proposal.
  - This project will happen no matter what, but with XLabs funding, we could truly change the world

#### Bibliography

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